

Technology News

Self-driving car

Google is working on a car that can drive by itself. Using artificial intelligence, the car can not only sense anything nearby but also mimic the decisions made by a human driver. The device atop the car produces a detailed map of the environment that is used for safe navigation.

The Internet giant has tested seven similar vehicles, which drove 1609 km (1000 miles) without human intervention and more than 225,308 km (140,000 miles) with only occasional human control. If estimates made by Google are anything to go by, self-driving vehicles are about eight years down the road. "Our goal is to help prevent traffic accidents, free up people's time and reduce carbon emissions by fundamentally changing car use," Google said on its blog.

The car can be programmed for different driving personalities, from cautious, in which it is more likely to yield to another car, to aggressive, where it is more likely to go first.

Sleepy-driver detector

Germany-based Fraunhofer Institute for Digital Media Technology is working on a driver-assist safety system that detects when the driver gets drowsy and warns with an audio alert. Equipped with up to six dashboard-mounted cameras having compact 3.4-mm lenses, the Eyetracker system can process up to 200 images per second to detect sleepiness using parameters like line of vision and eyelid position, irrespective of the spatial position of the driver's head.

The information is computed by a matchbox-sized processor that can be retrofitted to any vehicle without the need for any external computer.

Peter Husar, a researcher at Fraunhofer labs, explains, "Since the Eyetracker is fitted with at least two



The Integrated Eyetracker developed by the Fraunhofer IDMT permits contact-free and completely calibration-free 3-D measurement of a person's line of vision (image courtesy: Fraunhofer IDMT)

cameras that record images stereoscopically—meaning in three dimensions—the system can easily identify the spatial position of the pupil and the line of vision."

Breakthrough in solar cell efficiency

DEK Solar has unveiled the results of a breakthrough research on the print-on-print (PoP) technique—an approach that improves cell efficiencies, can be used in mainstream commercial production and can reduce the costs of production.

The research characterises the variables within an optimised PoP process and offers clear, concise descriptions of how to effect this within existing printing processes. One of the only routes to higher-aspect-ratio silver conductors in a volume manufacturing environment, PoP enables cell manufacturers to print higher, narrower collector grid features onto the front of solar cells. This, in turn, allows more sunlight through to the silicon, optimises the grid's electri-

cal performance and delivers high efficiency energy conversion capabilities that are key to future success.

Since PoP enables the metallisation grid to be printed onto the solar cell twice, repeatable accuracy is the enabler. Even the slightest misalignment can result in wider features and a reduction in cell efficiency.

Computers running without electricity

Researchers at the Ohio State University are working on a project to run computers using their own waste heat, not electricity. They are trying hard to develop a semiconductor called gallium manganese arsenide, which could make it possible to operate computers on heat. According to team leaders Joseph Heremans and Roberto Myers, "The research undertaken blends two cutting-edge technologies: thermoelectricity and spintronics."

The computers will recycle part of their own waste heat using gallium manganese arsenide. The researchers